

04 • L2 — High-level workflow (BPMN)

Assignment: “High-level workflow: diagrammatic representation of the workflow using BPMN.”

4.1 Overview

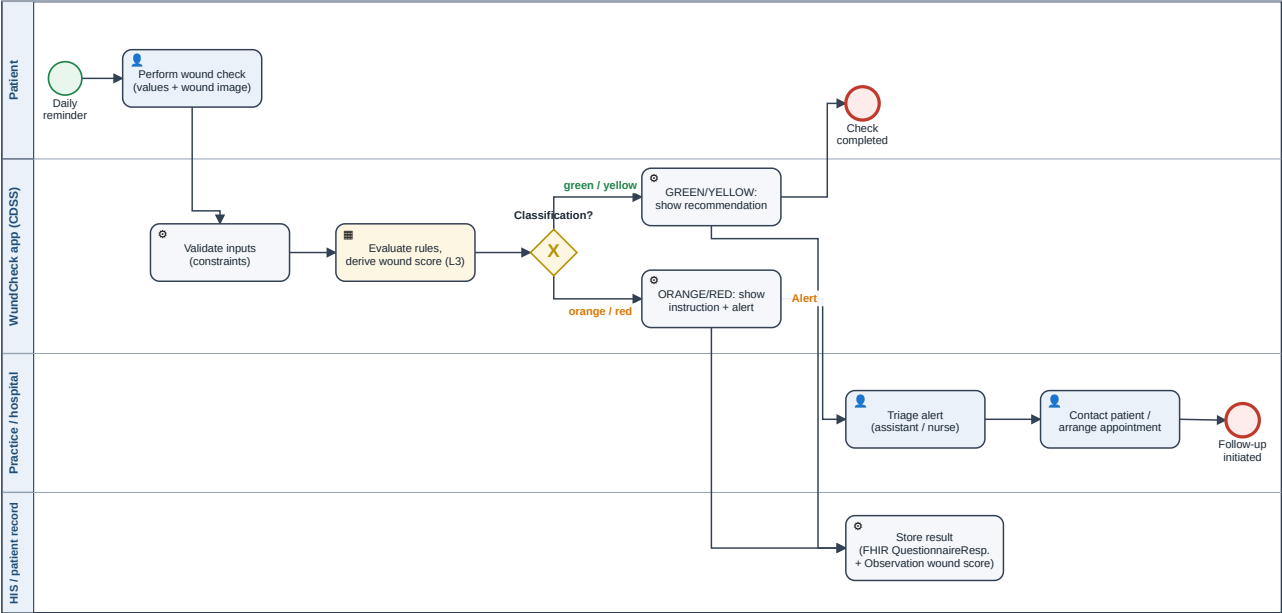
We document the workflow on **two levels**:

Diagram	Purpose	File
High-level workflow	Interaction of all parties involved — what happens where, who does what	diagrams/workflow-highlevel.bpmn
App detail process	What happens inside the app — validation and rule cascade	diagrams/workflow-app-detail.bpmn

Both are genuine BPMN 2.0 files, editable in bpmn.io and Camunda Modeler — not a redrawn flowchart. .svg and .png versions are provided alongside for embedding in presentations.

4.2 High-level workflow

WundCheck — High-level workflow (BPMN): daily wound check after surgery with primary closure



Legend: User Task • Service Task (automated) ■ Business Rule Task (decision table L3) • X = exclusive gateway — editable source: workflow-highlevel.bpmn (bpmn.io / Camunda Modeler)

Lanes

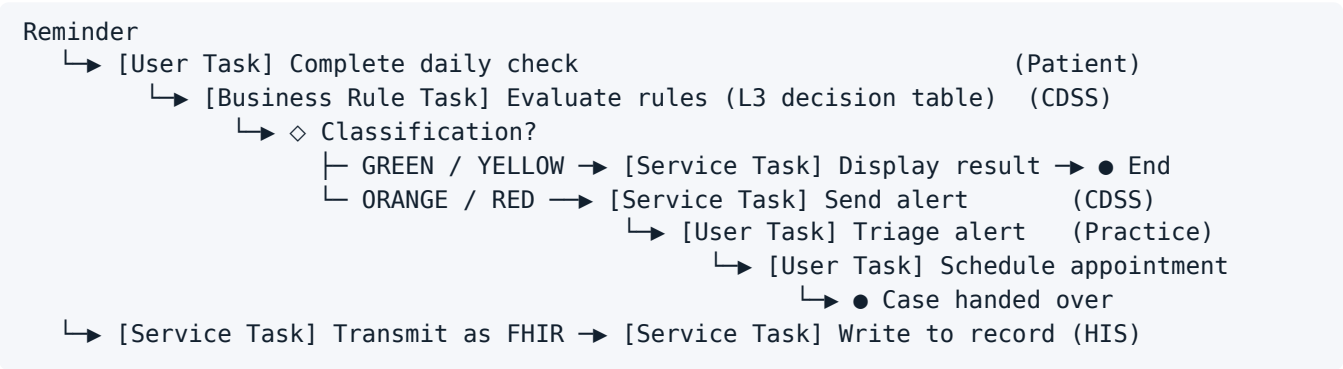
Lane	Responsibility
Patient	Recording the daily check, acting on the recommendation
WundCheck app (CDSS)	Validation, rule evaluation, classification, transmission
Practice / clinic	Receiving and processing alerts, scheduling appointments

Lane	Responsibility
HIS / patient record (EHR)	Persisting the history, making it available for the follow-up consultation

BPMN elements used

Element	Count	Where used
Start Event	1	Daily reminder to the patient
User Task	3	Activities involving humans (complete the check, process the alert, schedule an appointment)
Service Task	4	Automated system actions (FHIR transmission, alert dispatch, persistence)
Business Rule Task	1	Core element: "Evaluate rules" — execution of the decision table from the L3
Exclusive Gateway	1	"Classification?" — branches into the four traffic-light paths
End Event	2	Check completed (green/yellow) · case handed over to the practice (orange/red)
Sequence Flow	12	All connections, labelled at the gateways

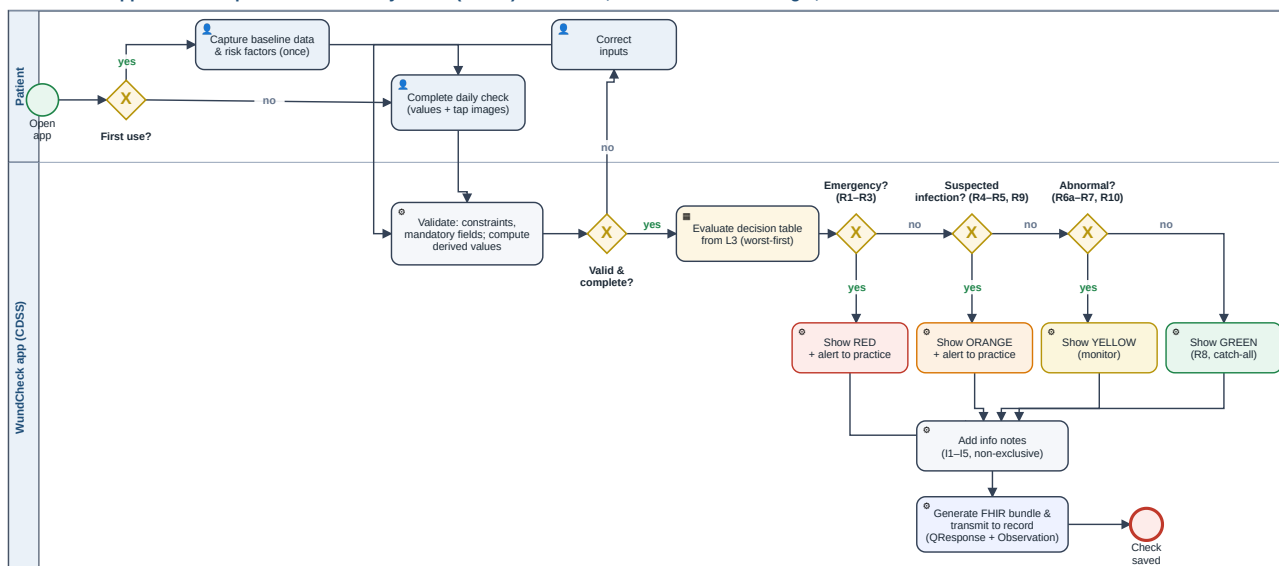
Process flow



Key design decision: Transmission to the record takes place on **every** completed run — including GREEN. Without this baseline, Dr Keller would only see the abnormal days at the suture-removal appointment and would be unable to assess the course over time (see Persona 3).

4.3 App detail process

WundCheck app — detailed process of one daily check (BPMN): validation, worst-first decision logic, FHIR transmission



Gateway cascade = decision logic in worst-first order (safety principle); emergency before suspected infection before abnormal; GREEN only as catch-all. Editable source: workflow-app-detail.bpmn

This diagram represents the **decision logic diagrammatically** and thereby complements the decision table from 06. The assignment permits either form — we deliver both, because they answer different questions: the table states *what* is decided, the diagram shows *in which order*.

Two core mechanisms

1 • Validation loop Before the rules are evaluated, a gateway checks **Valid & complete?**. If a constraint is violated (e.g. a temperature of 45 °C) or a mandatory entry is missing, the process returns to the input step — with the error message from the `constraint_message` of the respective data element. Data quality is thus enforced **before** rule evaluation rather than repaired afterwards.

2 • Worst-first gateway cascade Classification runs as a chain of exclusive gateways in descending order of severity:

Gateway	Rules checked	Result on a match
Emergency? (R1–R3)	Systemic signs, lymphangitis, dehiscence/pus	● RED — the cascade ends
Suspected infection? (R4–R5, R9, R11–R12)	Purulent/malodorous discharge, fever, feeling unwell, persistent discharge, deterioration	● ORANGE
Abnormal? (R6a–R7, R10)	Signs of inflammation, wound gap, bleeding under anticoagulation	● YELLOW
(no match)	Catch-all R8	● GREEN

In addition, a gateway **First use?** determines whether the one-off baseline data (surgery details, risk factors) must be recorded or can be skipped — a pure UX optimisation that keeps the daily time on task below 3 minutes (NFR-3).

Why worst-first?

For a triage tool in the hands of patients, **false reassurance** weighs more heavily than a false alarm. The cascade guarantees that, when several rules apply simultaneously, the most severe one always wins — regardless of how many reassuring findings sit alongside it. A patient with a scab (information rule I4, reassuring) **and** purulent discharge (R4, ORANGE) receives ORANGE.

Tested in T10 and T15.

4.4 Data objects

The detail process specifies `dataInput` elements at the relevant tasks. They make explicit which data elements feed into which step — and thereby establish the link to the Data Dictionary.

Task	Input data
Record baseline data	<code>pat_*</code> , <code>op_*</code> , risk factors
Record general condition	<code>temp</code> , <code>pain</code> , <code>pain_trend</code> , <code>wellbeing</code> , <code>chills</code> , <code>nausea_vomiting</code>
Assess the wound	<code>wound_state</code> , <code>redness_extent</code> , <code>swelling</code> , <code>secretion</code> , <code>bleeding</code> , <code>wound_open</code> , <code>warmth</code> , <code>red_streak</code>
Evaluate rules	all collected + calculated elements

4.5 Corrections made

Verified against `diagrams/workflow-app-detail.bpmn` :

- ☒ The pool in the detail diagram was named `Pool Participant` ; it is now "WundCheck app (CDSS)"
- ☒ The 7 generic `<bpmn:task>` elements are now typed as user or service tasks — zero untyped tasks remain
- ☒ The gateway labels reference the current rule set: `Emergency? (R1–R3) · Suspected infection? (R9, R4–R5, R11–R12) · Abnormal? (R6a–R7, R10)`
- ☒ Both diagrams are in English; `.svg` and `.png` re-rendered from source

Related documents: 05 Data Dictionary · 06 Decision Logic · 09 L4 & UX